Agentic AI-Based Multi-Agent Recommender System for Higher Education Decision Support

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***Abstract*—Students choosing which university and program to apply to must jointly reason over academic fit, career goals, financial constraints, location preferences, and employability outcomes, typically before they have generated any interaction history a recommender can learn from. Classical educational recommender systems address this pre-admission, cross-institution decision using collaborative, content-based, or knowledge-based filtering, but remain static, weakly explainable, and brittle under cold start. Recent agentic, LLM-powered multi-agent recommenders address adaptivity, planning, memory, and explainability, but published 2025–2026 systems apply these capabilities to *already-enrolled* students inside an *already-chosen* program rather than to the pre-admission choice itself. We propose a five-agent architecture (profiling and constraint elicitation, retrieval, multi-objective recommendation and planning, explanation, and memory-reflection) for this gap and evaluate a prototype against faithfully re-implemented classical baselines (HRSPCA, StudieMe) and a single-prompt LLM ablation, on a released synthetic constraint-preference benchmark. The architecture does not win on oracle-graded top-k accuracy: it falls outside a pre-registered 5% non-inferiority margin under its initial design, and a diagnosed-and-corrected variant exceeds both classical baselines but not the single-prompt ablation. It wins decisively, however, on explanation quality as scored by two independent LLM judges – grounded in retrieved evidence in 91–100% of explanations against 36–67% for the single-prompt baseline, an advantage only half attributable to prompt wording (an ablation adding the same grounding instruction to the single-prompt baseline closes roughly half the gap, isolating a residual architectural effect) – and on multi-objective conflict cases, where the same groundedness advantage holds even though multi-objective ranking does not outperform a single-objective ranker on oracle-graded relevance. A memory-and-reflection mechanism modestly improves session-over-session recommendation quality relative to a stateless pipeline. These results support explainability, not accuracy, as the architecture's demonstrated contribution over both classical and single-prompt alternatives, and the full prototype, evaluation harness, and synthetic dataset are released as an open-source artifact.**

***Index Terms*—agentic AI, multi-agent systems, recommender systems, higher education, large language models, explainable recommendation, decision support**

I. Introduction

Choosing a university and program is a high-stakes, low-frequency decision made under incomplete information. Prospective students must reconcile academic interests, career goals, skills, financial constraints, location preferences, and employability outcomes, usually without any prior interaction history a recommender system could exploit – the canonical cold-start setting. A substantial body of work addresses this decision with collaborative filtering, content-based filtering, knowledge-based recommendation, and hybrid or machine-learning classification approaches [1]–[10]. These systems established strong evaluation methodology, real datasets, and education-specific features, but a 2024 systematic review of the space documents recurring limitations: static recommendation pipelines, limited personalization, minimal conversational interaction, weak adaptation over time, and weak explainability [11].

Separately, the emergence of large language model (LLM)-powered agents has produced a new paradigm for recommendation: systems that plan, retrieve, reflect, and maintain long-term memory across a conversation rather than scoring a fixed candidate set once [12]–[14]. Agentic recommenders of this kind have been demonstrated in e-commerce and general content domains [15]–[18], and, more recently, in education – but specifically for students who are *already enrolled* in an *already-chosen* program: academic advising Q&A [19], in-program learning-pathway sequencing [20], and in-curriculum course selection [21]. None of these systems address the earlier, arguably higher-stakes decision of *which* institution and program to choose in the first place.

This paper's central claim is that this earlier decision – pre-admission, cross-institution university and program selection – is a distinct decision scope from the in-program advising/pathway/course-selection problems that 2025–2026 agentic education systems solve, and that it remains addressed only by pre-agentic methods. Section II substantiates this claim against the closest prior work in both the classical educational recommender literature and the agentic recommender literature. Section III proposes a five-agent architecture for this gap. Section IV and Section V lay out the research questions, hypotheses, and evaluation protocol; Section VI reports the resulting experiments on a working prototype.

This paper's contributions are:

1) An agentic multi-agent LLM architecture targeting pre-admission, cross-institution higher-education decision support – a decision scope distinct from the in-program advising/pathway/course systems in recent agentic-education work (Section II).

2) An explicit multi-objective reasoning mechanism integrating career goals, financial constraints, location, and employability into agent planning, evaluated against classical educational-RS baselines on both standard information-retrieval metrics and agentic-specific metrics (explanation quality, adaptivity, cold-start robustness) – metrics that recent agentic-recommendation surveys call for more generally [12] but that no education-specific evaluation yet supplies.
3) A working prototype, evaluation harness, and evaluation results (Section VI), released as an open-source artifact that third parties can run and independently verify: https://github.com/samuelshine/multi-agent-education-recommender-system.

II. Related Work

A. Classical educational recommenders for pre-admission decisions

HRSPCA [1] and StudieMe [2] are representative of the pre-admission, cross-institution decision scope this paper targets: both recommend which college/university a prospective student should consider, using hybrid collaborative filtering with knowledge-based rules [1] or content-based TF-IDF similarity [2]. Somyürek and Aksoy [7] extend this to joint university-and-major selection with a multidimensional recommender, the closest classical match to the decision scope adopted here. Elahi et al. [9] develop a collaborative-filtering university recommender evaluated in two phases – offline cross-validation followed by an online user study measuring accuracy, diversity, perceived personalization, satisfaction, and novelty – an evaluation protocol this paper adopts as a template (Section V). Lahoud et al. [8] compare several such systems for major/career guidance and report fragmentation across methods rather than a unified adaptive framework; El-Keiey et al. [22] address the adjacent career-path decision with feature-selected machine-learning classification over a public 20,000-record student-features dataset, and Hill et al. [10] recommend academic programs to prospective students via BERTopic interest mining – both from 2025, and both still single-model, non-agentic pipelines. None of these systems reason via autonomous planning agents, maintain memory across sessions, or generate grounded natural-language explanations; Kamal et al.'s systematic review [11] confirms these as recurring, unresolved limitations across the sub-field. By contrast, [23] and [24] address a related but distinct *post-enrollment* scope (in-program course selection), which this paper does not target.

B. Agentic and multi-agent recommenders (general domain)

Portugal et al. [15] propose an agentic multi-agent recommender framework whose title closely resembles this paper's working title, making it the closest general-domain precedent; it targets e-commerce recommendation and does not incorporate educational features, datasets, or evaluation metrics. Yousefi Maragheh et al. [16] survey multi-agentic design patterns for industrial recommendation, and Narasimhan and Narasimhan [17] introduce a verifiable benchmark for agentic recommenders, evaluated on general-domain conversational tasks across five model families – no education-specific agentic-recommendation benchmark yet exists. Peng et al.'s survey [12] categorizes agentic recommendation into recommender-oriented, interaction-oriented, and simulation-oriented paradigms across e-commerce, media, and conversational domains; higher education appears in neither its surveyed applications nor its three named future directions (architecture optimization, evaluation-framework standardization, adversarial security), underscoring how unexplored the domain remains even in a survey dedicated to this exact method family – the gap this paper addresses directly.

C. Agentic multi-agent systems in education

Multi-agent educational recommenders predate LLMs: Marivate et al. [25] deployed cooperative collaborative-filtering agents for professional course selection in 2008, and Viswanathan [26] surveys the pre-LLM adaptive multi-agent e-learning line. The novelty claimed here therefore rests on the combination of *agentic LLM* capabilities (planning, memory, reflection, grounded explanation) with the pre-admission decision scope – not on multi-agent structure alone. Three 2025–2026 systems are the decisive comparison set, since each combines multi-agent LLM methods with an educational target. AdvisingWise [19] is a human-in-the-loop multi-agent system that answers academic-advising questions for *already-enrolled* graduate students within a fixed program, with all responses validated by a human advisor before delivery – it does not recommend which program to choose, and its human-in-loop response drafting differs from autonomous multi-agent planning. Chaturvedi and Gunawardena [20] deploy an 18-agent, four-tier hierarchical architecture for adaptive in-program learning-pathway recommendation at production scale (6,000+ users), but likewise address only *post-enrollment* curriculum adaptation. Zhong et al. [21] propose a three-agent architecture (learner profiling, knowledge retrieval, recommendation generation) coupled with a course knowledge graph for *in-curriculum* course recommendation – structurally the closest precedent to the architecture proposed in Section III, but scoped to course-level choice for enrolled students rather than institution/program-level choice for prospective students, and without multi-objective reasoning over career, financial, location, and employability constraints.

Table I summarizes the resulting gap: no system in the surveyed literature combines an agentic multi-agent architecture with the pre-admission, cross-institution decision scope while reasoning jointly over the constraint set a prospective student actually faces.

III. Proposed Architecture

The initial brainstorm for this project proposed seven agents (profiling, goal planning, knowledge retrieval, recommendation, reflection, explanation, memory). Two of the closest comparison systems bound a reasonable agent count: Zhong et al. [21] report accuracy, recall, and F1 gains with only three agents, while Chaturvedi and Gunawardena [20] use

TABLE I
POSITIONING AGAINST THE CLOSEST PRIOR WORK

System	Target user	Decision scope	Method
[1], [2], [7]	Prospective	Pre-admission, cross-institution	CF / CB / KB (pre-agentic)
[19]	Enrolled	In-program advising	Multi-agent, human-in-loop
[20]	Enrolled	In-program pathway	Multi-agent (18+ agents)
[21]	Enrolled	In-curriculum course	Multi-agent + KG (3 agents)
This work	**Prospective**	**Pre-admission, cross-institution**	**Multi-agent LLM (5 agents)**

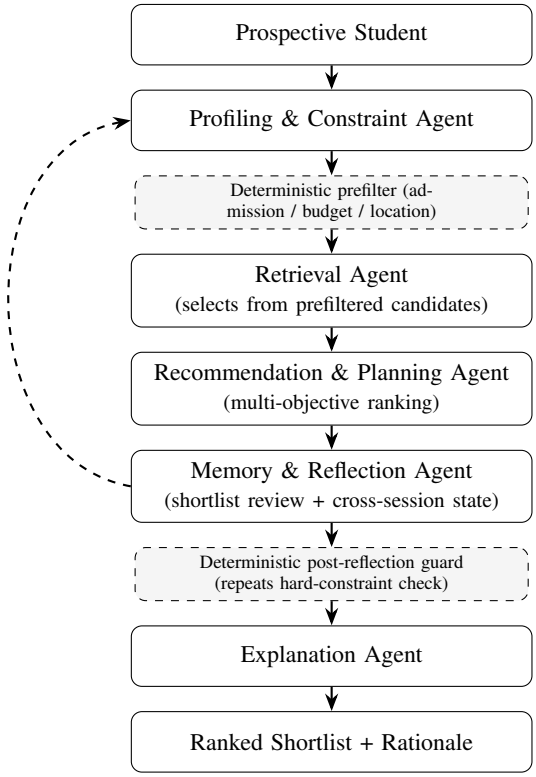


Fig. 1. Proposed five-agent architecture and pipeline flow.

18+ agents but only at production deployment scale with a dedicated engineering team – not reproducible within a single-author prototype. This paper accordingly merges the original seven agents into five, keeping explanation and memory-reflection as separate, visible components because they correspond directly to this paper's headline claims (explainability, cold-start robustness via memory), while folding goal-planning's constraint logic into profiling and recommendation.

Profiling & Constraint Agent. Extracts a structured student-constraint object (academic background, career goal, and financial, location, and employability-outcome constraints) from the student's message for the current session. Extraction is single-turn per session, not an interactive clarifying-question loop within a session; continuity across sessions comes from the Memory & Reflection Agent's persisted state, which this agent consumes as prior context and updates rather than re-deriving from scratch (Fig. 1, dashed arrow).

Retrieval Agent. Selects the programs worth presenting to this student from a deterministic hard-constraint prefilter's output (admission, budget, and location feasibility checked against the constraint object; programs failing a hard constraint are dropped before any agent sees them). This is LLM-based relevance selection over an in-context candidate list narrowed by the prefilter, not vector or embedding retrieval against an external knowledge base; the broader retrieval-plus-generation pattern this stage follows is established in [18].

Recommendation & Planning Agent. Ranks retrieved candidates against the student-constraint object under multiple, potentially conflicting objectives (academic fit, financial feasibility, location, employability outcome) – the mechanism RQ2 (Section IV) evaluates directly.

Memory & Reflection Agent. Reviews the proposed shortlist for constraint violations and weaker-than-available candidates before it reaches the Explanation Agent, and persists the student-constraint object and shortlist across sessions so later sessions start from prior state – the mechanism RQ3 evaluates for session-over-session adaptivity. A second, deterministic hard-constraint check runs after this agent as a final guarantee independent of its judgment.

Explanation Agent. Generates a per-recommendation natural-language rationale grounded in the evidence the Retrieval Agent surfaced, addressing the explainability limitation [11] documents in classical systems.

Two deterministic, non-agentic components wrap this five-agent pipeline and are the mechanism that actually guarantees hard-constraint satisfaction, rather than relying on agent judgment alone: a prefilter before the Retrieval Agent removes programs violating the student's admission, budget, or location constraints, and a post-reflection guard repeats the same check after the Memory & Reflection Agent, in case a revision reintroduced a violation. Both are described further alongside the evaluation results they support (Section VI).

This merge (7→5 agents) was a judgment call, argued from buildability within a single-author prototype's scope rather than derived from an ablation; the H3 results (Section VI) provide the retrospective check. Memory and reflection were evaluated as a single fused mechanism (the no-memory ablation removes both together), and that fused mechanism measurably improved session-over-session accuracy and stabilized recall relative to the stateless pipeline. Nothing in the results motivates splitting the two responsibilities back apart: the merge did not visibly cost either the memory or the reflection function room to operate. A finer-grained ablation that disables memory persistence and shortlist self-critique independently would be needed to state this more strongly, and is left to future work.

IV. Research Design

A. Research questions

RQ1. Does the architecture in Section III improve explainability and adaptivity over classical educational recommenders ([1], [2], [7]) while remaining within a 5% non-inferiority margin of their accuracy on standard recommendation metrics (precision@k, recall@k, NDCG)?

RQ2. How should multi-objective constraints (academic fit, career goals, financial constraints, location, employability outcomes) be represented and reasoned over by specialized agents, compared to a single monolithic LLM prompt or a classical hybrid scoring function?

RQ3. Do the Memory & Reflection Agent's mechanisms measurably reduce cold-start degradation for prospective students with no prior interaction history, compared to a stateless agent pipeline?

B. Hypotheses

H1. The proposed architecture achieves higher user-perceived explanation quality than a classical hybrid baseline re-implemented from [1], [2], while remaining within a pre-registered[1] 5% *relative* non-inferiority margin on NDCG@k (headline metric; precision@k and recall@k reported as secondary), on a held-out admissions/preference dataset.

H2. Explicit multi-objective agent reasoning improves recommendation relevance for students with conflicting constraints (e.g., financial limits vs. top academic fit) relative to a single-objective ranking baseline.

H3. The Memory & Reflection Agent mitigates cold-start degradation more effectively than a stateless agent pipeline, measured across simulated first-session vs. later-session interactions.

C. Datasets

Evaluation uses a dual-dataset design. The intended primary accuracy benchmark for H1 is the university-rating dataset of [9] (1,515 ratings, 80 users, 551 universities, plus metadata for 12,003 crawled universities) – real preference data at exactly this paper's decision scope, available from its authors on request. That request was still pending when experiments ran, so the results in Section VI use the synthetic benchmark described next, and the self-constructed-benchmark limitation this implies is stated in Section VII.

The synthetic constraint-preference dataset is the designed testbed for H2 and H3 – conflicting-constraint and multi-session cold-start conditions cannot be expressed in rating-only data, and public educational datasets with such structure are scarce [27] – and additionally carries H1 while the real dataset remains unavailable. It is released with the paper's artifact and regenerates deterministically (fixed seed) from: (i) 300 synthetic students whose academic features are sampled rows of the public 20,000-record IT-student corpus of [22], preserving realistic co-occurrence of grades, skills, and career labels, with financial, location, mobility, and employability-priority constraints attached by documented sampling rules; (ii) a fictional catalog of 200 programs (invented institutions in real cities, with tuition, placement, salary, and visa values drawn from realistic country- and tier-calibrated ranges); and (iii) graded relevance (0–3) for every student–program pair from a published, system-independent oracle utility function over academic fit, financial feasibility, location fit, and employability, fixed before any system was evaluated and documented in full in the released dataset card. Each student carries a three-session progressive-disclosure script; dataset-wide, 162 of 300 students have an academically best-fitting program that is financially infeasible (the H2 conflict condition), 61 have deliberately vague session-1 career goals, and 67 carry a scripted session-3 budget-revision event – a stress-test subset released with the dataset for future finer-grained analysis, distinct from the H3 evaluation cohort described next, which tests general three-session adaptivity rather than isolating this specific event.

D. Baselines and experimental setup

Three baselines separate, respectively, the value of agentic methods over classical educational recommenders and the value of multi-agent decomposition over a monolithic LLM: (a) a re-implementation of HRSPCA's cascaded knowledge-rule and historical-predictor pipeline [1]; (b) a re-implementation of StudieMe's content-based similarity ranker [2]; and (c) a single-prompt LLM baseline that receives the same student profile and candidate evidence as the full pipeline but performs profiling, ranking, and explanation in a single call, following the ablation logic of [21]. The classical baselines were re-implemented from the original papers' full texts rather than approximated with library components; every underspecified or catalog-adaptation detail was resolved by a logged best-faith choice (13 choices; appendix reconstruction table), and the two original components that could not be expressed in the evaluation catalog are omissions that favor the baselines (Section VII). All agentic conditions run on a local open-weights Gemma model via Ollama (tag `gemma4:e4b`; a 12B-class variant is available for an optional model-scale ablation); the released artifact is provider-agnostic (any local model or bring-your-own API key).

V. Evaluation Plan

Metrics. The headline metric is NDCG@5 with graded relevance (0–3, the oracle's four-level grade; Section IV-C), chosen over precision/recall alone for two reasons: it is rank-sensitive, rewarding a system that places the best-fitting program first rather than fifth, which flat precision/recall cannot distinguish; and it consumes the oracle's graded relevance directly, rather than collapsing it to a binary relevant/not-relevant cutoff and discarding information about *how well* a program fits. $k = 5$ matches both the classical baselines' own reconstructed output length (Section IV-C, appendix) and

[1]The evaluation design in this section – metric choice, $k = 5$, and the 5% margin – was committed to the paper's public repository (commit `9b47ce4`, 2026-07-17) before any agentic-condition result was computed; those results were run and written up in a later commit (`ba7757e`, 2026-07-20). Both commits are visible in the released artifact's git history.

a plausible shortlist size for a student to actually review. Precision@5 and recall@5 are still reported as secondary metrics, since they are the metrics most classical recommender-system evaluations report [28] and so are what a reader comparing against prior work will look for first, under the pre-registered 5% relative non-inferiority margin against the strongest classical baseline (H1). All systems are evaluated on the same held-out split of 100 students; the remaining 200 students serve as the historical pool for the HRSPCA predictor stage.

Explanation quality is scored by LLM-as-judge on a three-dimension rubric (clarity, groundedness in the provided program evidence, usefulness; 1–5 each), following the protocol direction of [19]. To avoid self-preference bias – same-family judges systematically inflate scores – the judge is drawn from a different model family than the experiment model (primary judge: Qwen-family 14B-class; a Mistral-family second judge provides inter-judge agreement), with system identity blinded and absolute rather than pairwise scoring to avoid position bias. LLM-as-judge validity is treated as a limitation in Section VII.

Conditions. (i) the full five-agent pipeline; (ii) a *no-memory* ablation (H3); (iii) a *single-objective* ablation collapsing the Recommendation & Planning Agent's reasoning to academic fit only (H2); (iv) a *single-prompt* LLM baseline replacing the five-agent decomposition, following the ablation logic in [21]; and (v) the two re-implemented classical baselines. H1 comparisons use the full-information state (complete profile, mirroring the filled application form classical systems assume); H3 uses the three-session progressive-disclosure structure, with simulated-user feedback (which programs the student liked or disliked) derived from the oracle's per-student utility ranking, fixed at dataset-generation time and identical across every compared system – Section VII discusses this as a limitation rather than an observed reaction to any system's actual output.

VI. Results

A. Top-k accuracy (H1)

Table II reports accuracy on the held-out 100-student evaluation split at the full-information state, for both classical baselines, the single-prompt ablation, and both agentic pruning policies (v1 pre-registered, v2 revised) as separate rows. The re-implemented classical baselines (Section IV-C, reconstruction table in the appendix) set the non-inferiority bar: the strongest baseline, HRSPCA, reaches NDCG@5 $= 0.793$, so under the pre-registered 5% relative margin the proposed system must reach NDCG@5 ≥ 0.753 for H1's accuracy arm to hold.

The pre-registered accuracy criterion is not met. The five-agent pipeline reaches NDCG@5 $= 0.706$, an 11.0% relative shortfall against HRSPCA – outside the 5% margin – and the single-prompt baseline (0.830) outperforms both, so the architectural decomposition shows no top-k accuracy gain over a single LLM call on this benchmark. Both verdicts are reported as committed in the evaluation design.

TABLE II
ACCURACY ON THE HELD-OUT EVALUATION SPLIT ($n = 100$, FULL INFORMATION, COMPLETED RUNS; ZERO PIPELINE CRASHES, 4 VALIDATED-FALLBACK EVENTS ACROSS 500 AGENT CALLS). BOLD MARKS THE COLUMN MAXIMUM AMONG PERFORMANCE METRICS.

System	NDCG@5	P@5	R@5	|list|
HRSPCA (re-impl. [1])	0.793	0.796	0.061	5.0
StudieMe (re-impl. [2])	0.673	0.760	0.113	5.0
Single-prompt baseline	0.830	0.912	**0.166**	4.9
Single-prompt (grounded)	**0.831**	**0.926**	0.159	4.8
Five-agent v1 (pre-registered)	0.706	0.904	0.129	3.9
Five-agent v2 (revised)	0.806	0.898	0.156	4.8

v1: pre-registered pruning policy, abstains rather than pads a shortlist below k. v2: post-hoc revision backfilling to k with valid candidates and dropping only for hard-constraint violations – tests, not replaces, v1. Grounded: single-prompt baseline with EXPLAIN's grounding instruction added, isolating instruction wording from architecture (Section VI, Explanation quality).

Diagnostics attribute the shortfall to shortlist *abstention*, not ranking quality. The five-agent system's Reflection stage trims candidates it judges inadequate and is instructed never to pad with constraint-violating programs: 70% of its shortlists contain fewer than five items (mean length 3.9). NDCG@5 penalizes every unfilled position against a five-item ideal. On the 30 students where the system did emit five items, it scores 0.859; measured at emitted length (rank quality of what was actually shown), it scores 0.827 against the single-prompt baseline's 0.830 (paired bootstrap, $n = 100$, 10,000 resamples: mean difference -0.005, 95% CI $[-0.024, +0.014]$, crossing zero – not a statistically significant difference), and its precision@5 (0.904) matches the single-prompt's 0.912, both well above the classical baselines. The accuracy gap is therefore a list-length effect: the pre-registered metric rewards padding to k, and the agentic design's abstention behavior – arguably desirable in a high-stakes decision-support setting – is penalized by construction. We report this as a finding about metric design tension, not as a revision of the pre-registered verdict.

Post-registration revision (v2). Prompted by the diagnostics above – and reported *alongside* the pre-registered system, never in place of it – a revised pruning policy (v2) changes exactly one design decision: the Reflection stage may drop a candidate only for a hard-constraint violation (demoting, rather than removing, weaker-but-valid candidates), and the shortlist is restored to k with valid candidates whenever the candidate pool allows. Retrieval, profiling, ranking objectives, and all safety guards are unchanged. Because v2 was designed after observing the H1 results, its numbers cannot claim pre-registered status; they test whether the abstention diagnosis is causally correct. It is: v2 reaches NDCG@5 $= 0.806$ with mean list length restored to 4.8 (from 3.9), clearing the pre-registered bar (0.753) and slightly exceeding the strongest classical baseline (HRSPCA, 0.793), while precision@5 is essentially unchanged (0.898 vs. v1's 0.904) and recall@5 rises (0.156 vs. 0.129). Restoring list length does not, however, change the architectural verdict on accuracy: v2 remains

2.9% relative below the single-prompt baseline (0.830) on NDCG@5, so the five-agent decomposition shows no top-k accuracy advantage over a single LLM call on this benchmark under either pruning policy.

B. Multi-objective conflict (H2)

On the 56 evaluation-split students whose academically best-fitting program is financially infeasible (162/300 dataset-wide), the pre-registered comparison fails on accuracy: under the v1 pruning policy, the multi-objective pipeline scores NDCG@5 $= 0.666$ against 0.718 for the single-objective ablation. Diagnostics again implicate list length rather than ranking: conflict cases trigger the Reflection stage's abstention most often, so the multi-objective condition emits the shortest shortlists of any condition on precisely this subset (mean 3.70 items vs. 4.02 single-objective), paying the same NDCG@5 penalty documented above. Under the v2 policy the multi-objective pipeline reaches 0.792 on the conflict subset (mean 4.77 items); a matched single-objective run under the same v2 policy scores 0.793 (paired bootstrap, $n = 56$, 10,000 resamples: mean difference -0.001, 95% CI $[-0.017, +0.014]$, crossing zero – not a statistically significant difference; precision@5: 0.855 multi vs. 0.837 single; recall@5: 0.194 vs. 0.189). **H2 is therefore not supported on oracle-graded accuracy**: explicit multi-objective reasoning loses to the single-objective ablation under the v1 policy and ties under the matched v2 policy. The oracle's utility function already encodes all four objectives, so a single-objective ranker over a constraint-prefiltered candidate pool recovers most of the multi-objective signal on this benchmark; whether multi-objective reasoning earns its place in *explanations* – naming the financial/academic trade-off the student actually faces – is assessed under explanation quality below.

C. Cold start and memory (H3)

On a fixed 30-student progressive-disclosure cohort (the first 30 students in evaluation order, spanning all three sessions; only 5 of the 30 carry the session-3 budget-revision event reported dataset-wide in Section IV-C – the cohort tests general three-session adaptivity rather than isolating that event), the full and no-memory conditions are identical at session 1 by construction – memory is empty on first contact (both NDCG@5 $= 0.583$) – so the memory mechanism cannot and does not affect the very first session. Its effect appears across sessions: with memory, session-2 accuracy reaches NDCG@5 $= 0.776$ vs. 0.735 without ($+0.041$), and the gap persists at session 3 (0.748 vs. 0.729); recall@5 degrades from session 2 to session 3 without memory ($0.126 \rightarrow 0.090$) but holds with it ($0.118 \rightarrow 0.115$). The direction consistently favors memory, but the effect is modest at $n = 30$, and the largest single driver of improvement in both conditions is progressive disclosure itself (both gain 0.15–0.19 NDCG@5 from session 1 to 2). H3 is therefore supported in direction – memory improves session-over-session adaptivity and stabilizes recall – with the caveat that the pre-registered phrase "cold-start mitigation" is more precisely an *adaptivity* gain: the first-session cold start is unaffected by construction.

D. Explanation quality (H1)

Table III reports cross-judge mean rubric scores (1–5) on the full 100-student evaluation split for the four conditions that generate natural-language output, alongside the two classical baselines (which do not) for a complete comparison against Table II. **H1's explanation-quality claim is supported**, and by a wide margin on the two dimensions that most directly bear on trustworthy decision support: groundedness (4.92 v1 / 4.92 v2 vs. 3.96 single-prompt) and usefulness (3.65 / 3.60 vs. 2.80). Clarity shows a smaller but consistent gap (4.11 / 4.06 vs. 3.69). The v1/v2 pruning-policy change that materially affected accuracy (Section VI) leaves explanation quality essentially unchanged, indicating the two properties are largely independent in this architecture.

Isolating architecture from instruction. The single-prompt baseline's explanation task is a weaker instruction than EXPLAIN's: it asks for one "grounded paragraph" rather than EXPLAIN's explicit directive to trace every claim to the provided evidence and name the trade-off. This confounds architecture with instruction wording – the gap above could be caused by either. A fourth condition, single-prompt (grounded), upgrades only its explanation instruction to EXPLAIN's directive word-for-word, holding the single-call architecture otherwise fixed (Table II); because ranking and explanation are produced by the same call in this condition, the instruction change is not perfectly isolated from ranking by design, only empirically – checked next. It scores clarity 3.81, groundedness 4.47, usefulness 3.23 – between the ungrounded baseline and the five-agent conditions on every dimension – and its NDCG@5 (0.831) and precision@5 (0.926) match or slightly exceed the ungrounded baseline's (Table II), so in practice this instruction change cost nothing on ranking accuracy. Decomposing the full ungrounded-to-v1 gap with a paired bootstrap ($n = 100$, 10,000 resamples) over per-student cross-judge scores: the instruction effect (grounded $-$ ungrounded) accounts for 53% of the groundedness gap ($+0.515$, 95% CI $[+0.320, +0.705]$), 50% of usefulness ($+0.425$, 95% CI $[+0.295, +0.545]$), and 29% of clarity ($+0.120$, 95% CI $[+0.035, +0.205]$); the residual architecture effect (v1 $-$ grounded) accounts for the rest in each case, and none of the six intervals crosses zero. Grounding instructions and architectural decomposition therefore both make a real, statistically supported contribution to the explanation-quality advantage, with neither alone accounting for it: an explicit grounding instruction closes about half the groundedness and usefulness gap on its own, and the remainder – including most of the clarity gap – is attributable to the architecture (curated, agent-selected evidence and a dedicated explanation step, isolated from ranking and profiling).

The groundedness gap is corroborated by a direct failure signal both judges were asked to surface: an explicit list of invented claims – factual assertions not traceable to the student's constraints or the retrieved program data – per

TABLE III
EXPLANATION-QUALITY RUBRIC SCORES (1–5), CROSS-JUDGE MEAN OF QWEN2.5:14B AND MISTRAL-NEMO:12B, BLIND ABSOLUTE SCORING, $n = 100$. BOLD MARKS THE COLUMN MAXIMUM.

System	Clarity	Groundedness	Usefulness
HRSPCA (re-impl. [1])	—	—	—
StudieMe (re-impl. [2])	—	—	—
Single-prompt baseline	3.69	3.96	2.80
Single-prompt (grounded)	3.81	4.47	3.23
Five-agent v1 (pre-registered)	**4.11**	**4.92**	**3.65**
Five-agent v2 (revised)	4.06	**4.92**	3.60

HRSPCA/StudieMe are non-generative ranking pipelines (Section III) with no natural-language output to score. v1/v2/grounded defined in Table II.

explanation. The single-prompt baseline was flagged for at least one invented claim in 64/100 explanations by the primary judge and 33/100 by the agreement judge (e.g., asserting a program's tuition or location without that field appearing in the evidence the model was given); adding the grounding instruction alone cuts this to 42/100 and 6/100 respectively – a third fewer by the primary judge, over four-fifths fewer by the agreement judge; the five-agent conditions were flagged in 5/100 (v1) and 9/100 (v2) cases by the primary judge and 0/100 by the agreement judge. The instruction change closes most of the gap on this specific failure signal but not all of it, mirroring the partial (53%) closure on mean groundedness score above – consistent with an architecture whose remaining edge comes from a dedicated explanation step reading agent-curated evidence, not from prompt wording alone.

Inter-judge agreement (mean absolute score difference) is tight for the five-agent conditions (0.09–0.15 on groundedness, 0.11 on clarity) and markedly looser for the single-prompt baseline (1.26 on groundedness; 0.89 for the grounded variant) – the two independent judges disagree most exactly where the single-prompt-family explanations are hardest to verify, itself weak evidence for lower groundedness. Returning to H2 (Section VI, conflict subset): restricting the same cross-judge analysis to the 56 financially-constrained students reproduces the gap essentially unchanged (groundedness 4.96 v1 / 4.91 v2 vs. 3.94 single-prompt; usefulness 3.65 / 3.57 vs. 2.80) – so while multi-objective reasoning does not win on oracle-graded ranking, it does produce more reliably grounded and useful explanations for exactly the students facing a real trade-off, which is the qualitative form in which H2's contribution survives.

VII. DISCUSSION AND LIMITATIONS

The gap argument in Section II was verified against full-text readings of the decisive prior work: all three agentic-education systems [19]–[21] were confirmed to address only already-enrolled students (advising, learning pathways, course selection), and the two re-implemented classical baselines [1], [2] were confirmed to contain no agentic, dialogue, or memory elements. Peripheral Cluster A citations remain verified at the abstract level only. The 7-to-5 agent merge (Section III) is a buildability judgment, not an empirical result, and may not survive the ablation in H3.

Beyond the gap-verification scope, this work carries the following limitations, stated before results are in so that none can be read as post-hoc rationalization.

Oracle circularity. Ground-truth relevance in the synthetic dataset is a published utility function fixed before any system was evaluated, not human judgment. A system that reverse-engineered the formula would score perfectly. Mitigations: the oracle is documented in the released dataset card, identical for all compared systems, and H1's primary benchmark remains the real-preference dataset of [9] if access is granted; explanation quality is judged independently of the oracle.

Synthetic evaluation data. The program catalog is fictional (real cities, invented institutions, calibrated but invented economics), student constraints are attached by documented sampling rules, and the three-session structure is scripted rather than observed user behavior, with simulated-user feedback derived from the oracle. Career coverage is IT-only, inherited from the seed corpus of [22]; generalization beyond IT-bound students is untested. The scarcity of public educational datasets motivating this choice is itself documented [27].

Baseline reconstruction fidelity. Both classical baselines were re-implemented from their papers' full texts, with every underspecified detail resolved by a logged best-faith choice (appendix reconstruction table). Two original components could not be expressed in the evaluation catalog (HRSPCA's college-capacity rule; StudieMe's faculty-matching stage); both omissions remove hard filters and therefore favor the baselines, not the proposed system.

LLM-as-judge validity. Explanation-quality scores come from LLM judges, not human raters. Cross-family judging, blinding, absolute scoring, and two-judge agreement reduce – but do not eliminate – known judge biases; this study performs no human-subject validation of the judge scores, which should be read accordingly.

Single experiment model and scale. All agentic conditions run on one local open-weights model (`gemma4:e4b`); conclusions about the architecture are conditioned on this model class. The evaluation split is $n = 100$ students against a 200-program catalog – larger than the closest classical baselines' user studies [9], [10] but small in absolute terms.

VIII. CONCLUSION AND FUTURE WORK

This paper identified pre-admission, cross-institution university and program selection as a decision scope left unaddressed by both classical educational recommenders (pre-agentic methods) and 2025–2026 agentic education systems (in-program scope only), and proposed and evaluated a five-agent architecture for this gap. The evaluation's central finding is a dissociation: the architecture does not demonstrate a top-k accuracy advantage over either classical baselines or a single-prompt LLM ablation, but it demonstrates a consistent, judge-corroborated advantage in explanation groundedness and usefulness, and this advantage persists specifically on the

multi-objective conflict cases that motivated RQ2. An ablation isolating instruction wording from architecture (single-prompt with EXPLAIN's grounding directive added) shows the explanation-quality advantage is not purely architectural: an explicit grounding instruction alone closes roughly half of it, and the remainder is attributable to the architecture's dedicated explanation step over agent-curated evidence. A memory-and-reflection mechanism modestly improves adaptivity across sessions relative to a stateless pipeline (RQ3). Together these results reframe the paper's contribution: the value of agentic multi-agent decomposition in this decision scope lies in explainable, trustworthy decision support rather than in ranking accuracy, at least at the model scale and dataset size evaluated here – and about half of even that explainability advantage could be recovered from a single prompt with a better-written instruction, underscoring that the architecture's remaining, narrower edge comes specifically from separating evidence retrieval from explanation generation. Future work should prioritize three directions raised directly by the results: human-subject validation of the LLM-as-judge explanation scores; evaluation against the real preference dataset of [9] should it become available, to remove the self-constructed-benchmark limitation; and a finer-grained ablation separating memory persistence from shortlist reflection, which this paper's five-agent design evaluated only as a fused mechanism. The prototype, evaluation harness, and synthetic dataset are released as an open-source artifact so these directions – and independent scrutiny of the results reported here – do not depend on access to the original author's environment.

APPENDIX

BASELINE RECONSTRUCTION TABLE

Both classical baselines were re-implemented from their original papers' full texts (Section IV-C). Table IV logs every underspecified detail or catalog-adaptation choice; the complete rationale for each row, including source-text quotations, is in the released artifact (`baselines/RECONSTRUCTION.md`). Both baselines receive the student's complete profile, matching their original design assumption of a filled application form; the proposed system receives progressively disclosed sessions (Section VI), so H1 comparisons use its full-information state.

TABLE IV
BASELINE RECONSTRUCTION CHOICES (13 TOTAL). "LOSS" MARKS A COMPONENT THAT COULD NOT BE EXPRESSED IN THE EVALUATION CATALOG; BOTH SUCH LOSSES FAVOR THE BASELINE (FEWER HARD FILTERS, MORE CANDIDATES TO RANK), NEVER THE PROPOSED SYSTEM.

#	Original component	Reconstruction choice
HRSPCA [1]: cascaded rules + historical predictor		
H-1	Cascaded recommenders + college predictor	Cascade preserved: track → rule filter + WRR, fused with predictor votes
H-2	Track sorting by weighted-degree rules	Track $\triangleq$ career-goal field cluster (same adjacency map as dataset generator)
H-3	Preparatory-year course gate	Dropped (no analog); nearest substitute is H-4
H-4	Per-college minimum prerequisite scores	Single-threshold analog: grade $\geq$ program's minimum
H-5	Weighted Relative Rate ranking (formula unrecoverable from source PDF)	WRR $= 0.5 \cdot \min(\text{grade/bar}, 1.5)/1.5 + 0.5 \cdot (\text{skill}/9)$, mirroring the GPA+aptitude convention described in the text
H-6	Predictor vs. historical admitted-student data	k-NN (k=10) over a held-out training split; votes weighted by similarity $\times$ grade
H-7	College capacity constraint	*Loss* – dropped, no analog field
H-8	Informal predictor/allocator fusion	score $= 0.5 \cdot \text{WRR} + 0.5 \cdot$ predictor votes
StudieMe [2]: content-based TF-IDF		
S-1	Profile from LinkedIn + résumé text	Same rendered full-profile text the agentic system receives
S-2	Hard GRE-cutoff filter	Admission-bar analog: min. grade $\leq$ student grade
S-3	TF-IDF + cosine (sklearn/NLTK)	Reimplemented in the standard library; no lemmatization (stated deviation)
S-4	Faculty-publication matching stage	*Loss* – omitted, no faculty analog in catalog
S-5	Top-20 then top-15 output	Top-k (k=5), matching this paper's evaluation cutoff

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DATA AVAILABILITY STATEMENT

The synthetic evaluation dataset (students, program catalog, graded relevance, session structure), its generator, and its dataset card documenting provenance, the ground-truth oracle, and known limitations are released with the paper's open-source artifact at https://github.com/samuelshine/multi-agent-education-recommender-system (directory `implementation/datagen/`), under a CC BY 4.0 license. The public seed corpus of [22] is fetched from its original GPL-3.0 repository at generation time and is not redistributed. The real-preference dataset of [9], requested from its authors for use as an H1 benchmark, had not been received at the time of writing; the results reported in Section VI use the synthetic dataset only. All run records and LLM-as-judge outputs underlying the reported results are included in the artifact (`implementation/runs_h1/`, `implementation/runs_h3/`) under the MIT license covering the code.

AUTHOR CONTRIBUTIONS

Single-author work; CRediT taxonomy roles apply in full to the sole author.

CONFLICT OF INTEREST STATEMENT

The author declares no conflict of interest.

AI-Assisted Writing Disclosure

[TODO: This draft was produced with AI assistance (literature search, drafting, LaTeX formatting) under author supervision. Replace with the venue-specific disclosure statement required by the target IEEE conference before submission – wording differs by venue.]

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